Algebraic Geometry

Shing Tak Lam

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Lecture 1

Preliminaries

Plan

Roughly speaking, there are four parts of the course:

1. Spectrum of a commutative ring, and basics of sheaf theory.
2. Definitions of schemes and morphisms between them.
3. Properties of schemes, and properties of morphisms (replacements of naïve topological notions of compactness, Hausdorff, etc.). Basic constructions (Picard group, class group).
4. Rapid introduction to sheaf cohomology.

Prerequisites

(i) Basic undegraduate mathematics (groups, rings, topological spaces and so on).

(ii) Commutative algebra (co-requisite).

Resources

(i) Dhruv's course page.

(ii) Textbooks: Hartshorne, Vakil for Algebraic geometry, Atiyah-MacDonald for commutative algebra.

Why suffer through this? (Weil conjectures)

The Weil conjectures link two worlds. The starting data is a homogeneous polynomial $f \in \mathbb{Z}[x_0, \dots, x_n] = \mathbb{Z}[\mathbf{x}]$, of degree d.

World 1: We can consider the vanishing locus

$$X = \mathbb{V}(f) = \{(a_0 : \cdots : a_n) \mid f(a_0, \dots, a_n) = 0\} \subseteq \mathbb{P}^1_{\mathbb{C}}$$

which is a projective hypersurface. We will assume that X is smooth, i.e. there is no point $p \in X$, such that all of the partial derivatives $\frac{\partial f}{\partial x_i}$ vanish for all i. If we view $\mathbb{P}^n$ as $(\mathbb{C}^{n+1} \setminus 0)/\text{scaling}$, and $\mathbb{C}^{n+1} \setminus 0$ is given the Euclidean topology, then X is a (complex) manifold. This follows from the inverse/implicit function theorem. Moreover, X is a compact topological space. In fact, X can be triangulated.

With this, we can consider various invariants associated to X. The simplest is the Euler characteristic $\chi(X)$. For example, $\chi(\mathbb{P}^k) = k + 1$. More generally, we can consider the Betti numbers b_i, where

$$b_i = \dim_{\mathbb{Q}} H_i(X; \mathbb{Q})$$

are the dimension of the rational homology groups. In this case,

$$\chi(X) = \sum_i (-1)^i b_i$$

World 2: Fix a prime p, such that X is smooth over $\overline{\mathbb{F}_p}$ (exercise: p exists). Define N_m to be the number of points in $X\left(\mathbb{F}_{p^m}\right)$.

We can package the data of the N_ms: Consider the formal power series

$$\zeta(X,t) = \exp\left(\sum_{m=1}^{\infty} \frac{N_m}{m} t^m\right)$$

which is the *Weil ζ function*.

Theorem 0.1 (Grothendieck).

1. $\zeta(X,t)$ is a rational functon. That is,

$$\zeta(X,t) = \frac{P_0(t)\cdots P_{2n-2}(t)}{P_1(t)\cdots P_{2n-3}(t)}$$

   where each $P_i(t)$ is a polynomial.

2. The degree of the polynomial $P_i(t)$ is $b_i(X)$.

Deligne proved the Riemann Hypothesis for the Weil ζ function.

1 Beyond Algebraic Varieties

1.1 Summary of affine varieties

Let $k = \overline{k}$ be an algebraically closed field. We would like to consider subsets

$$\mathbb{V}(S) \subseteq \mathbb{A}_k^n$$

where $S \subseteq k[x_1, \ldots, x_n]$. Up to isomorphism, this is the same as finitely generated k-algebras without nilpotent elements. Given $V = \mathbb{V}(S)$, we have an ideal

$$I(V) = \{f \in k[x_1, \ldots, x_n] \mid f = 0 \text{ on } V\}$$

We can then define

$$k[V] = \frac{k[x_1, \ldots, x_n]}{I(V)}$$

Conversely, given $k[V]$, a finitely generated nilpotent-free k-algebra. Choosing generators $y_1, \ldots, y_n$ of the algebra, we have a surjection $\pi : k[x_1, \ldots, x_n] \twoheadrightarrow k[V]$, sending x_i to y_i. We can then take

$$V = \mathbb{V}(\ker(\pi))$$

Lecture 2

Let $\sqrt{I}$ be the radical of an ideal I be

$$\sqrt{I} = \{f \in k[\mathbf{x}] \mid f^m \in I \text{ for some } m\}$$

Then $\mathbb{V}(S) = \mathbb{V}(\langle S\rangle) = \mathbb{V}(\sqrt{\langle S\rangle})$, where $\langle S\rangle$ is the ideal generated by S. In this case, we have a correspondence between affine varieties in $\mathbb{A}_k^n$ and radical ideals in $k[\mathbf{x}]$.

Remark 1.1. This awkwardness caused by "in affine space" can be removed, by considering varieties up to isomorphism, and coordinate rings (i.e. finitely generated nilpotent free k-algebras).

1.1.1 Basic structures

- $k[\mathbf{x}]$ is the ring of algebraic functions on $\mathbb{A}_k^n$.
- $I \subseteq k[\mathbf{x}]$ is a radical ideal.
- The coordinate ring of $V = \mathbb{V}(I)$ is $k[V] = k[\mathbf{x}]/I$.

This coordinate ring is the basic object, it's more fundamental than I.

1.1.2 Topology

Fix an affine variety $V \subseteq \mathbb{A}_k^n$. It has a Zariski topology, where the closed sets are subsets of the form

$$V \cap \mathbb{V}(S)$$

where $S \subseteq k[\mathbf{x}]$. Equivalently, we can consider the Zariski topology on $\mathbb{A}_k^n$, and then equip V with the subspace topology.

Intrinsically, we can say that $W \subseteq V$ is closed if we have that

$$W = \mathbb{V}(J)$$

where $J \subseteq k[V]$ is an ideal. Exercise: these two topologies are the same.

1.1.3 Nullstellensatz

That is, what are the maximal ideals in $k[\mathbf{x}]$? Equivalently, in a nilpotent free finitely generated k-algebra $k[V]$?

Fix $V = \mathbb{V}(I)$, and pick $p \in V$. There is a natural ring homomorphism $\mathrm{ev}_p : k[V] \to k$, given by $\mathrm{ev}_p(f) = f(p)$.

Claim 1.2. ev_p is surjective.

Proof. Exercise. □

The kernel of ev_p is a maximal ideal in $k[V]$. Therefore, points on V are one source of maximal ideals.

Theorem 1.3 (Nullstellensatz)**.** There is a bijection

$$\{\text{points in } V\} \leftrightarrow \{\text{maximal ideals in } k[V]\}$$

1.1.4 Functions

Each $f \in k[V]$ gives a function $V \to k(= \mathbb{A}_k^1)$, sending p to $f(p)$. Equivalently, if we denote $\mathfrak{m}_p = \ker(\mathrm{ev}_p)$, then this is the quotient map

$$k[V] \twoheadrightarrow k = k[V]/\mathfrak{m}_p$$

1.1.5 Morphisms

Given $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$, morphisms $V \to W$ are constructed as follows.

1. Take functions $f_1, \ldots, f_m \in k[x_1, \ldots, x_n]$.
2. This gives a map $\varphi = (f_1, \ldots, f_m) : V \to \mathbb{A}_k^m$.
3. A morphism $V \to W$ must additionally satisfy that $\varphi(V) \subseteq W$.

There is a more intrinsic construction. Any such $\varphi : V \to W$ as above, gives rise to a pullback map $\varphi^* : k[W] \to k[V]$, where $\varphi^*(g) = g \circ \varphi$.

There is a bijection

$$\{\text{Morphisms } V \to W\} \leftrightarrow \{k\text{-algebra homomorphisms } k[W] \to k[V]\}$$

In practice, the right hand side is much cleaner than the right hand side.

Lesson: We ♡ coordinate rings.

Upshot: Let R be a finitely generated, nilpotent free k-algebra, the topological space

$$X_R = \{\mathfrak{m} \subseteq R \mid \mathfrak{m} \text{ is a maximal ideal}\}$$

Closed sets are given by $\mathbb{V}(S)$, for $S \subseteq R$.

1.2 Limitations

Question 1.4. What is an abstract (not necessarily affine) variety?

Morally, it's an object which locally looks like an affine variety.

Example 1.5 (Non-algebraically closed fields)
Consider the ideal

$$I = \left\langle x^2 + y^2 + 1 \right\rangle \in \mathbb{R}[x, y]$$

Then $\mathbb{V}(I) = \varnothing \subseteq \mathbb{R}^2$. I is a prime ideal, therefore radical, so the naïve Nullstellensatz fails.

Question 1.6. Given a commutative ring R, is there a natural topological space X_R, on which R is the ring of functions? For example, $R = \mathbb{Z}, \mathbb{Z}[x]$ and so on.

Example 1.7 (Why restrict to radical ideals)
Assume $k = \overline{k}$.

$$C = \mathbb{V}(y - x^2) \subseteq \mathbb{A}_k^n$$

and

$$D = \mathbb{V}(y)$$

Then

$$C \cap D = \mathbb{V}(y, y - x^2) = \mathbb{V}(x^2, y) = \{(0, 0)\}$$

But $\left\langle y, y - x^2 \right\rangle$ has radical $\langle x, y \rangle$, it is not equal to the ideal $\langle x, y \rangle$. The intersection wants to be two points. Take

$$D_\delta = \mathbb{V}(y + \delta)$$

where $\delta \in k \setminus 0$, then $C \cap D_\delta$ has two points.

1.3 Spectrum of a ring

Basic idea: Given a ring A, we will define

$$\operatorname{Spec}(A) = \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ a prime ideal}\}$$

Why prime?

- Functoriality - $A \to B$ induces $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ by pullback.
- Some rings have very few maximal ideals.

Lecture 3

Let A be a commutative ring,
Goal: Define a topological space X_A on which A is naturally the ring of functions.

Definition 1.8 (Zariski spectrum)
The Zariski spectrum of A is

$$\operatorname{Spec}(A) = \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ a prime ideal}\}$$

Remark 1.9. We might also consider the maximal ideal spectrum, but this fails to be functorial. That is, a ring homomorphism $\varphi : A \to B$ induces a map $\varphi^* = \varphi^{-1} : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$, by considering the preimage. On the other hand, no such map exists for maximal ideals.

Why? Whatever X_A could be, given a point $p \in X_A$, we should have an evaluation map

$$\begin{aligned} \mathrm{ev}_p : A &\to K_p \\ f &\mapsto f(p) \end{aligned}$$

where K_p is a field. Note we allow the field K_p to depend on p. Nor are we requiring that ev_p be surjective. The image of ev_p is an integral domain, and moreover, any map to an integral domain R is also a map to a field $F = \mathrm{Frac}(R)$.

Subject to this, homomorphisms $f : A \to K$, with K a field, corresponds to $\ker(f)$, which is a prime ideal.

Example 1.10

Taking $A = \mathbb{Z}$, we have that

$$\mathrm{Spec}(\mathbb{Z}) = \{\langle p \rangle \mid p \in \mathbb{N} \text{ a prime}\} \cup \{0\}$$

Pick a function $123 \in \mathbb{Z}$. Now evaluate at different points, which is just reduction modulo p, i.e.

$$\begin{aligned} 123(2) &= 1 \in \mathbb{Z}/2 \\ 123(3) &= 0 \in \mathbb{Z}/3 \\ 123(0) &= 123 \in \mathbb{Z} \subseteq \mathbb{Q} \end{aligned}$$

Example 1.11

Taking $A = \mathbb{R}[x]$, $\mathrm{Spec}(A)$ has elements corresponding to

1. irreducible quadratics in x, e.g. $\langle x^2 + 1 \rangle$, which corresponds to $(\mathbb{C}\backslash\mathbb{R})$/complex conjugation.
2. linear polynomials in x, e.g. $\langle x - 2 \rangle$, corresponding to $\mathbb{R}$,
3. 0.

Hence $\mathrm{Spec}(A)$ is the upper half plane, plus the point zero.

Definition 1.12

If A is an integral domain, then $0 \in \mathrm{Spec}(A)$ is called the generic point.

Remark 1.13. The value of $f \in A$ at $\mathfrak{p} \in \mathrm{Spec}(A)$ is $f + \mathfrak{p} \in A/\mathfrak{p}$.

1.4 Topology on $\mathrm{Spec}(A)$

Motivation: In the Zariski topology, the zero sets should be the vanishing sets of functions. Fix $f \in A$, define the vanishing set of f to be

$$\mathbb{V}(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \in \mathfrak{p}\} = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f(\mathfrak{p}) = 0 \in A/\mathfrak{p}\}$$

More generally, if $I \subseteq A$ is an ideal, then

$$\mathbb{V}(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid I \subseteq \mathfrak{p}\} = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f(\mathfrak{p}) = 0 \text{ for all } f \in I\}$$

Definition 1.14

The sets $\mathbb{V}(I) \subseteq \operatorname{Spec}(A)$, for $I \subseteq A$ ideals, satisfy the requirements for closed sets in a topological space.

Proof. Easy to see $\varnothing$ and $\operatorname{Spec}(A)$ are closed. Moreover,

$$\bigcap_\alpha \mathbb{V}(I_\alpha) = \mathbb{V}\left(\sum_\alpha I_\alpha\right)$$

and

$$\mathbb{V}(I_1) \cup \mathbb{V}(I_2) = \mathbb{V}(I_1 \cap I_2)$$

For this, $\subseteq$ is clear. For the reverse containment,

$$I_1 I_2 \subseteq I_1 \cap I_2$$

If $\mathfrak{p} \supseteq I_1 \cap I_2$, then it contains $I_1 I_2$. Since $\mathfrak{p}$ is prime, then either I_1 or I_2 is contained in $\mathfrak{p}$. □

Example 1.15

Let $k = \overline{k}$, and $A = k[x, y]$. Then

$$\operatorname{Spec}(A) \supseteq \operatorname{maxSpec}(A) = k^2$$

where $\operatorname{maxSpec}(A)$ is the set of maximal ideals, which by the Nullstellensatz corresponds to the points in $\mathbb{A}^2_k$. There is a point for each non-maximal prime ideal, e.g.

$$\langle x - y \rangle \quad \left\langle y^2 - x^3 \right\rangle \quad \left\langle y - x^2 \right\rangle$$

Consider the generic point $0 \in \operatorname{Spec}(A)$. Then the closure of $\{0\}$ is $\operatorname{Spec}(A)$. On the other hand, the closure of a maximal ideal is itself.

Now consider the ideal $\langle f \rangle$, where f is an irreducible polynomial, e.g. $y^2 - x^3$. The closure of $\{\langle f \rangle\}$ contains every maximal ideal $\langle x - a, y - b \rangle$ such that $b^2 - a^3 = 0$, since such maximal ideals contain $\langle f \rangle$.

Consider the ring $A = \mathbb{C}[[x]]$, the ring of formal power series in x. Then

$$\operatorname{Spec}(A) = \{\langle 0 \rangle, \langle x \rangle\}$$

which

$$\overline{\{\langle 0 \rangle\}} = \operatorname{Spec}(A)$$
$$\overline{\{\langle x \rangle\}} = \{\langle x \rangle\}$$

On the other hand, $\operatorname{maxSpec}(A)$ is a single (closed) point, which is the same as $\operatorname{Spec}(\mathbb{C})$.